

Muhammad Munhib Baig

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Professional Summary

Graduating Computer Science engineer from FAST-NUCES (2026) with strong foundations across software engineering, AI, backend development, and cloud infrastructure. Demonstrated ability to deliver production-grade systems end-to-end: from architecture and implementation to deployment and maintenance. Eager to contribute across the stack in a team environment and grow into a senior engineering role.

Experience

Codify Labs

Software Developer Intern

July 2026 – Present

Lahore, Pakistan

- Developing Python automation scripts and internal tooling, applying system design principles and clean code practices in a professional environment.
- Designing modular, maintainable solutions adhering to industry-standard software engineering practices.

Technical Skills

- Languages:** Python, JavaScript, C++, C
- Backend:** Node.js, Express.js, FastAPI, REST APIs
- Frontend:** React
- AI / ML:** PyTorch, Scikit-learn, YOLOv8, OpenCV, LLMs, AI Agents, MCP
- Cloud & DevOps:** AWS EC2, DynamoDB, IAM, AWS CLI, Linux, SSH, PM2, Git
- Databases:** MongoDB, MySQL, SQLite, DynamoDB
- Data & Visualisation:** Pandas, NumPy, Power BI, Tableau

Projects

A.R.E.S — Autonomous Response & Emergency Surveillance (FYP)

2025–2026

Lead Engineer — Final Year Project

Python, YOLOv8, Flower, FastAPI, OpenCV, SQLite

- Led the design and implementation of a privacy-preserving AI surveillance system detecting criminal activities across distributed CCTV streams using **Federated Learning**.
- Improved model accuracy from **73.8% to 83.9%** through federated aggregation across four edge clients; built a real-time inference pipeline achieving **sub-two-second** alert latency.
- Integrated all components via a **FastAPI** backend with SQLite persistence and an Electron desktop frontend, demonstrating full-stack delivery capability.

MentorOS — AI Agent Platform

2026

Backend Contributor — Collaborative Project

Python, MCP, LLMs, Git

- Contributed backend components to a collaborative AI-powered development assistant, working within a structured codebase with strict quality tooling (Black, Ruff, Mypy, Pytest).
- Implemented agent orchestration, memory handling, and configuration management modules, gaining experience with professional software engineering processes and PR-based Git workflows.

Cloud-Native E-Commerce Platform (Browniebees)

2025

Full Stack / Cloud Engineer — Personal Project

Node.js, AWS EC2, DynamoDB, Linux

- Independently designed, developed, and deployed a complete e-commerce application on AWS: built the Node.js API, configured the cloud infrastructure, and managed production operations.
- Gained practical experience with IAM, security groups, SSH, PM2 process management, and Git-based deployment workflows.

Business Performance Dashboard

2024

Data Analyst — Personal Project

Power BI, Tableau, Excel

- Built interactive BI dashboards over a 10,000+ record retail dataset, practising end-to-end data analysis: cleaning, modelling, visualisation, and stakeholder-facing insight delivery.

Education

National University of Computer and Emerging Sciences (FAST-NUCES)

Aug 2022 – Jun 2026

Bachelor of Science in Computer Science

Relevant Coursework: Software Engineering, AI, Machine Learning, Cloud Computing, Database Systems, Web Technologies, Operating Systems, Computer Networks, Data Science